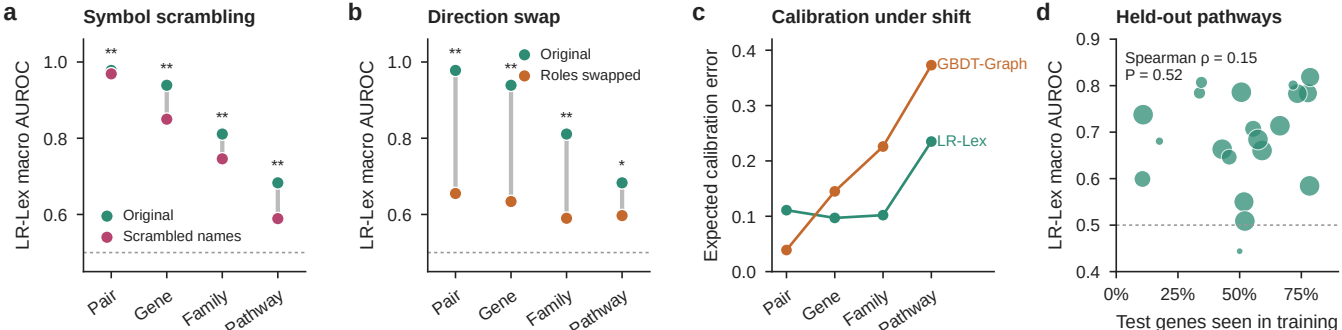
a
Symbol scrambling
LR-Lex macro AUROC
1.0
0.8
0.6
**
**
**
**
Original
Scrambled names
Pair
Gene
Family
Pathway
b
Direction swap
LR-Lex macro AUROC
1.0
0.8
0.6
**
**
**
*
Original
Roles swapped
Pair
Gene
Family
Pathway
c
Calibration under shift
Expected calibration error
0.4
0.3
0.2
0.1
0.0
GBDT-Graph
LR-Lex
Pair
Gene
Family
Pathway
d
Held-out pathways
LR-Lex macro AUROC
0.9
0.8
0.7
0.6
0.5
0.4
Spearman ρ = 0.15
P = 0.52
0%
25%
50%
75%
Test genes seen in training